

## Generalizing the Series

Consider the general infinite series:

$$\mathcal{S} = \sum_{n=1}^{\infty} \frac{1}{an^2 - an + b}$$

where  $a, b \in \mathbb{R}$  and  $a \neq 0$ .

First, we determine the roots of the quadratic denominator  $ax^2 - ax + b = 0$ :

$$r_{1,2} = \frac{a \pm \sqrt{a^2 - 4ab}}{2a} = \frac{1}{2} \pm \frac{\sqrt{a^2 - 4ab}}{2a}$$

A fundamental property of these roots is their symmetry around  $\frac{1}{2}$ , meaning  $r_1 + r_2 = 1$ , or  $r_2 = 1 - r_1$ . Let the difference between the roots be denoted by  $\Delta = r_1 - r_2 = \frac{\sqrt{a^2 - 4ab}}{a}$ .

Using partial fraction decomposition, the summand can be rewritten as:

$$\begin{aligned} \frac{1}{an^2 - an + b} &= \frac{1}{a(r_1 - r_2)} \left( \frac{1}{n - r_1} - \frac{1}{n - r_2} \right) \\ &= \frac{1}{a\Delta} \left( \frac{1}{n - r_1} - \frac{1}{n - (1 - r_1)} \right) \end{aligned}$$

This allows us to express the infinite sum in terms of the digamma function,  $\psi(z)$ , which is defined by the series representation:

$$\psi(z) = -\gamma + \sum_{n=0}^{\infty} \left( \frac{1}{n+1} - \frac{1}{n+z} \right)$$

For a telescoping-like sum of rational functions, the standard identity is:

$$\sum_{n=1}^{\infty} \left( \frac{1}{n - \alpha} - \frac{1}{n - \beta} \right) = \psi(1 - \beta) - \psi(1 - \alpha)$$

Substituting  $\alpha = r_1$  and  $\beta = 1 - r_1$  into the identity yields:

$$\mathcal{S} = \frac{1}{a\Delta} [\psi(1 - (1 - r_1)) - \psi(1 - r_1)] = \frac{1}{a\Delta} [\psi(r_1) - \psi(1 - r_1)]$$

By invoking Euler's reflection formula for the digamma function,  $\psi(1 - z) - \psi(z) = \pi \cot(\pi z)$ , we arrive at the exact closed-form generalization:

$$\mathcal{S} = -\frac{\pi \cot(\pi r_1)}{a\Delta}$$

## Evaluating the Specific Limit

Applying this analytical framework to the specific expression:

$$\lim_{k \rightarrow \infty} \sum_{n=1}^k \frac{1}{100n^2 - 100n + 9}$$

Here,  $a = 100$  and  $b = 9$ . We first isolate the roots of  $100x^2 - 100x + 9 = 0$ :

$$r_{1,2} = \frac{100 \pm \sqrt{10000 - 3600}}{200} = \frac{100 \pm 80}{200}$$

This gives rational roots  $r_1 = \frac{9}{10}$  and  $r_2 = \frac{1}{10}$ .

The difference between the roots is  $\Delta = 0.9 - 0.1 = 0.8$ .

Substituting these into our generalized formula:

$$\mathcal{S} = -\frac{\pi \cot(0.9\pi)}{100(0.8)} = -\frac{\pi \cot(\pi - 0.1\pi)}{80}$$

Using the periodicity and symmetry of the cotangent function,  $\cot(\pi - \theta) = -\cot(\theta)$ , this simplifies to:

$$\mathcal{S} = \frac{\pi \cot\left(\frac{\pi}{10}\right)}{80}$$

Recognizing the exact trigonometric constant  $\cot\left(\frac{\pi}{10}\right) = \sqrt{5 + 2\sqrt{5}}$ , the rigorous exact value of the limit is:

$$\lim_{k \rightarrow \infty} \sum_{n=1}^k \frac{1}{100n^2 - 100n + 9} = \frac{\pi \sqrt{5 + 2\sqrt{5}}}{80}$$